MAJOR ARTICLE

# Pharmacokinetics and Safety of Extemporaneously Compounded Valacyclovir Oral Suspension in Pediatric Patients from 1 Month through 11 Years of Age

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***Background.*** Valacyclovir provides enhanced acyclovir bioavailability in adults, but limited data are available in children.

***Methods.*** Children 1 month through 5 years of age with or at risk for herpesvirus infection received a single 25 mg/kg dose of extemporaneously compounded valacyclovir oral suspension ($n = 57$), whereas children 1 through 11 years of age received 10 mg/kg valacyclovir oral suspension twice daily for 3–5 days (herpes simplex virus infection) ($n = 28$) or 20 mg/kg 3 times daily for 5 days (varicella-zoster virus infection) ($n = 27$). Blood samples for pharmacokinetic analysis were collected during the 6 h after the first dose. Safety was monitored throughout the studies.

***Results.*** Dose proportionality in the maximum observed concentration ($C_{max}$) of acyclovir and the area under the concentration-time curve from time zero extrapolated to infinity ($AUC_{0-\infty}$) existed across the 10 to 20 mg/kg valacyclovir dose range. For children 2 through 5 years of age, an increase in dose from 20 to 25 mg/kg resulted in near doubling of the $C_{max}$ and $AUC_{0-\infty}$. Among infants 1 through 2 months of age receiving 25 mg/kg, the mean $AUC_{0-\infty}$ and $C_{max}$ were higher (~60% and 30%, respectively) than those among older infants and children receiving the same dose. Valacyclovir oral suspension was well tolerated. No clinically significant trends were noted in clinical chemical, hematologic, or urinalysis values from screening to follow-up.

***Conclusions.*** Among children 3 months through 11 years of age, the 20 mg/kg dose of this formulation of valacyclovir oral suspension produces favorable acyclovir blood concentrations and is well tolerated. A dosing recommendation cannot be made for infants <3 months of age because of decreased clearance in this age group.

***Trial registration.*** ClinicalTrials.gov identifier: NCT00297206.

Infections due to the α-subfamily of human herpesviruses (herpes simplex virus [HSV] and varicella-zoster virus [VZV]) are common during childhood. HSV type 1 (HSV-1) can cause asymptomatic infection or orolabial disease. When patients are symptomatic, the primary infection produces herpes gingivostomatitis, whereas recurrent infection causes herpes labialis, commonly referred to as *fever blisters* or *cold sores*. One-third of all 6–13-year-old children in the United States have had an HSV-1 infection, and among African American children the proportion is ~50% [1]. A total of 60%–90% of Americans have acquired HSV-1 by the age of 50 years [2]. Varicella historically has been an even more prevalent infection. Before licensure of the monovalent varicella vaccine in 1995, ~4 million cases of varicella occurred annually in the United States, reflecting near-universal infection during childhood [3, 4]. Even with high varicella vaccination rates, however, vaccine effectiveness for prevention of disease of any severity has been 80%–85% [5], and as a result large numbers of individual breakthrough varicella cases continue to occur [6].

Acyclovir therapy is indicated for life-threatening

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HSV and VZV infections, such as HSV encephalitis, neonatal HSV infections, and VZV infections in immunocompromised hosts [7–13]. Acyclovir therapy also improves the outcome of herpes labialis [14, 15], varicella [16, 17], and herpes zoster [18, 19]. The clinical utility of oral acyclovir, however, is limited by its low oral bioavailability and need for frequent dosing.

Valacyclovir is the L-valyl ester of acyclovir that is rapidly converted to acyclovir after oral administration by intestinal and liver valacyclovirase [20]. Its mechanism of action, antiviral spectrum, and resistance are the same as those of its parent drug, acyclovir [21]. With an oral bioavailability 3–5 times greater than oral acyclovir, valacyclovir can be administered less frequently than oral acyclovir and can achieve plasma concentrations similar to intravenous acyclovir in adults. Because attempts to develop a standard suspension formulation were not successful because of stability issues, an extemporaneous oral suspension formulation of valacyclovir, made from valacyclovir caplets just before dispensing, was developed by GlaxoSmithKline. This article presents pharmacokinetic and safety data on the use of this oral suspension formulation in children 1 month through 11 years of age.

## METHODS

***Study designs and populations.*** From January 2003 to February 2007, a total of 3 phase 1, open-label, multicenter studies were conducted at 15 sites in Australia, Chile, Estonia, South Africa, and the United States. Each study differed slightly in study design and population, but all had the primary objective of determining the pharmacokinetics and safety of the valacyclovir oral suspension formulation in children with or at risk for herpesvirus infections. Study 914 enrolled children 1 month through 5 years of age who had an active herpesvirus infection (eg, herpes labialis or varicella) or who were at risk of acquiring a herpesvirus infection (eg, patients at risk of acquiring herpes labialis or chickenpox or patients with a history of neonatal herpes or herpes labialis at risk of skin recurrences). Participants in studies 915 and 916 were 1 through 11 years of age and had clinical diagnoses of HSV and VZV infection, respectively. Participants in all 3 studies were required to have adequate renal function (creatinine clearance, >50 mL/min/1.73 $m^2$), as determined by the equation described by Schwartz et al [22]. Participants in each study were stratified by age: 1 through 2 months (study 914), 3 through 5 months (study 914), 6 through 11 months (study 914), 12 through 23 months (studies 914, 915, and 916), 2 through 5 years (studies 914, 915, and 916), and 6 through 11 years (studies 915 and 916).

The studies were conducted in accordance with good clinical practice guidelines and all applicable regulatory requirements, including the October 2000 version of the Declaration of Helsinki [23]. Before conducting any study-specific procedures, written informed consent was obtained from each participant's parent or legally authorized representative and, when appropriate, verbal assent was obtained from the child. Each child was seen on at least 2 occasions after the eligibility screening visit: day 1, at which dosing commenced, and 2–4 days after the last dose of study medication.

***Study drug preparation.*** Valacyclovir oral suspension was prepared in concentrations of 25 or 50 mg/mL in lots of 100 mL, as detailed in the US prescribing information [24].

***Study treatments.*** Doses of valacyclovir were selected to approximate the acyclovir exposures (area under the plasma concentration–time curve [AUC]) observed in previous studies in adults taking oral valacyclovir at approved adult doses. Doses according to body weight were chosen on the basis of considerations that acyclovir is eliminated predominantly as unchanged drug via the kidney and the expected age-related changes in renal function [25]. Dosing regimens evaluated were 25 mg/kg as a single dose (study 914, to approximate the acyclovir exposure from 1000 mg of valacyclovir in adults), 10 mg/kg as a twice-daily dose for 3–5 days (study 915, to approximate the acyclovir exposure from 500 mg of valacyclovir in adults and evaluate multidose safety and tolerability), and 20 mg/kg as a 3-times-daily dose for 5 days (study 916, to approximate the acyclovir exposure from 1000 mg of valacyclovir in adults and evaluate multidose safety and tolerability).

***Bioanalysis and pharmacokinetics.*** In all 3 studies, pharmacokinetic samples were to be collected from each child at predose and at 0.5, 1, 2, 4, and 6 h after the first dose. Acyclovir, valacyclovir, and their internal standards ($[^{13}C^{15}N_3]$-acyclovir and $[^{13}C^{15}N_3]$-valacyclovir, respectively) were isolated from human plasma by a solid-phase extraction procedure. Samples were then reconstituted and quantitated by high-performance liquid chromatography/tandem mass spectrometry using an Applied Biosystems/MDS SCIEX API 3000 system with TurboIonSpray in the positive ion mode. The method has a lower limit of quantitation of 10 ng/mL for acyclovir and valacyclovir using a 50-$\mu$L aliquot of human plasma. The validation range was 10–1000 ng/mL for both analytes. Within- and between-run precision and accuracy of quantitation was $\pm$11.6% for acyclovir and $\pm$6.9% for valacyclovir across the concentration range.

Acyclovir pharmacokinetic parameter estimates for each child were obtained by standard noncompartmental analysis using WinNonlin Professional Edition, version 4.1 (Pharsight). These estimates included maximum observed concentration ($C_{max}$), time to peak concentration ($t_{max}$), AUC extrapolated to infinity ($AUC_{0-\infty}$), elimination half-life ($t_{1/2}$), and apparent oral clearance (CL/F). The dose amount used for determination of CL/F was corrected for the difference in molecular weight between acyclovir (225 kDa) and valacyclovir (324 kDa) (CL/F = dose/AUC = 225/324 $\times$ [mg/kg valacyclovir dose]/acyclovir $AUC_{0-\infty}$]). All children for whom acyclovir pharmacokinetic



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**Table 1. Demographic and Baseline Characteristics of Participants in Studies 914, 915, and 916**

| Demographic or baseline characteristic | Study | | | All |
|---|---|---|---|---|
| | 914 | 915 | 916 | |
| Total no. of patients | 57 | 28 | 27 | 112 |
| Sex | | | | |
| Female | 25 (44) | 17 (61) | 14 (52) | 56 (50) |
| Male | 32 (56) | 11 (39) | 13 (48) | 56 (50) |
| Race | | | | |
| American Indian or Alaska Native | 9 (16) | 0 (0) | 0 (0) | 9 (8) |
| Asian | 0 (0) | 0 (0) | 1 (4) | 1 (1) |
| Black | 24 (42) | 6 (21) | 2 (7) | 32 (29) |
| White | 23 (40) | 15 (54) | 19 (70) | 57 (51) |
| Other | 1 (2) | 7 (25) | 5 (19) | 13 (12) |
| Age | | | | |
| 1 through 2 months | 14 (25) | … | … | 14 (13) |
| 3 through 5 months | 12 (21) | … | … | 12 (11) |
| 6 through 11 months | 10 (18) | … | … | 10 (9) |
| 12 through 23 months | 9 (16) | 7 (25) | 6 (22) | 22 (20) |
| 2 through 5 years | 12 (21) | 13 (46) | 13 (48) | 38 (34) |
| 6 through 11 years | … | 8 (29) | 8 (30) | 16 (14) |
| Height, mean cm ± SD | 73.4 ± 18.4 | 105.9 ± 25.7 | 103.0 ± 18.9 | NC |
| Weight, mean kg ± SD | 10.0 ± 6.2 | 19.8 ± 11 | 17.2 ± 6.4 | NC |

**NOTE.** Data are no. (%) of study participants, unless otherwise indicated. NC, not calculated; SD, standard deviation.

parameter estimates could be obtained were included in the pharmacokinetic analyses. Pharmacokinetic results were compared with historical data from previously conducted pediatric studies.

***Safety assessments.*** Blood chemical analyses, hematologic laboratory tests, and urinalyses were performed before dosing on day 1 and at the follow-up visit 2–4 days after dosing. Adverse events were monitored throughout each study. An adverse event was defined as any untoward medical occurrence in a patient temporally associated with the use of a medicinal product, whether or not it was considered related to the product. Clinically significant abnormal laboratory findings were reported as adverse events, unless they were present at screening and did not worsen significantly. At each visit, an analysis for thrombotic microangiopathy (TMA) was conducted. This analysis comprised an assessment of hemoglobin, platelets and schistocytes, renal abnormalities, neurologic abnormalities, and fever. All children who received at least 1 dose of valacyclovir oral suspension were included in the safety analyses.

***Sample size.*** The sample size for each study was based on the number of study participants considered adequate to characterize the pharmacokinetics of acyclovir after administration of valacyclovir oral suspension. At least 6 participants were enrolled into each age cohort in each study.

## RESULTS

***Study population.*** The studies enrolled 112 children from 1 month through 11 years of age, of whom 50% were male and 51% were white. Demographic details of study participants are provided in Table 1. A higher percentage of participants in study 914 were African American than in study 915 or 916. However, in study 914 no statistically significant differences in acyclovir $C_{max}$, AUC, or $t_{1/2}$ between these racial groups (black vs other) were seen ($P$ values from unpaired $t$ tests all >.36), and there is no theoretical rationale for a difference in valacyclovir absorption, conversion of valacyclovir to acyclovir, or acyclovir disposition between blacks and others. All 112 study participants were included in the safety analyses, and 98 participants had an adequate number of pharmacokinetic data points available to be included in the pharmacokinetic analyses.

All participants in study 914 received a single dose of valacyclovir suspension (25 mg/kg). Twenty-six of the 28 participants in study 915 completed the protocol-specified 3–5 days of treatment with valacyclovir (10 mg/kg twice daily); one participant missed the fifth dose of valacyclovir because of a medication spill, and another participant missed the sixth dose because of being asleep. Twenty-three of the 27 participants in study 916 completed the protocol-specified 5 days of valacyclovir treatment (20 mg/kg 3 times daily); 2 participants discontinued treatment prematurely (day 4 and day 5); 1 participant spit out the first dose and refused any further dosing; and 1 participant missed the fifth dose. The median duration of valacyclovir exposure was 3 days in study 915 (range, 1–6 days; mean, 3.9 days) and 5 days in study 916 (range, 1–7 days; mean, 5.1 days).

***Pharmacokinetics results.*** Valacyclovir was rapidly ab-

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sorbed after oral administration and extensively converted to acyclovir. As expected because of rapid conversion from prodrug to active drug, valacyclovir concentrations were low compared with acyclovir concentrations. Comparing geometric mean values for all study participants, valacyclovir $C_{max}$ values were ~3% of acyclovir $C_{max}$ values (molar ratio) (data not shown), similar to results in adults [26].

Acyclovir pharmacokinetic results are provided in Table 2 and Figures 1 and 2. For comparison, Table 2 and Figures 1 and 2 also contain pediatric data on pharmacokinetics after parenteral acyclovir administration from prior studies. Within-study comparisons of acyclovir mean results across pediatric age groups at both the 10 and 20 mg/kg doses of valacyclovir suggest that exposures are somewhat lower in the 2- through 5-year age group relative to children 12 through 23 months of age or 6 through 11 years of age. At the 25 mg/kg dose, a similar pattern of differing exposure among age groups was observed, with the lowest exposures at 6 months through 1 year of age, compared with infants 1 through 5 months of age and with children 2 through 5 years of age; older age groups were not included in the 25 mg/kg dose study (study 914). As expected, patients in age ranges with relatively lower exposures had a higher drug CL/F than did patients in age ranges with relatively higher exposures. Clearance data from studies of parenteral acyclovir showed a similar increase in clearance during the first few years of life relative to infants or older children (Figure 3).

Notably, in the youngest age group assessed at 25 mg/kg (infants 1 through 2 months of age), mean acyclovir exposures were higher (~60% and 30% higher for $AUC_{0-\infty}$ and $C_{max}$, respectively) compared with exposures in older infants and children at the same dose. In slightly older children 3 months through 1 year of age, exposures were modestly lower (20% and 24% lower for $AUC_{0-\infty}$ and $C_{max}$, respectively) than those

**Table 2. Mean Acyclovir Pharmacokinetic Parameters after First-Dose Administration of Valacyclovir Compared with Historical Studies of Intravenous Acyclovir**

| Patient age, treatment | Dose | Mean (%CV) | | | | Bioavailability estimate, % |
|---|---|---|---|---|---|---|
| | | $AUC_{0-\infty}$, μg·h/mL | $C_{max}$, μg/mL | $t_{1/2}$, h | CL/F or CL, mL/min/kg | |
| 1 through 2 months | | | | | | 45 |
| Oral valacyclovir ($n$ = 9) | 25 mg/kg | 27.6 (25) | 7.0 (22) | 2.0 (23) | 11.0 (25) | |
| Intravenous acyclovir ($n$ = 12) [7, 27] | 15.8 mg/kg[a] | 58.3 (47) | 18.6 (38) | 2.5 (34) | 5.0 (30) | |
| 3 through 5 months | | | | | | 22 |
| Oral valacyclovir ($n$ = 9) | 25 mg/kg | 17.7 (37) | 5.2 (43) | 1.8 (41) | 21.6 (85) | |
| Intravenous acyclovir ($n$ = 3) [27] | 14.2 mg/kg[a] | 50.5 (16) | 17.6 (16) | 2.8 (53) | 4.7 (16) | |
| 6 through 11 months | | | | | | UD |
| Oral valacyclovir ($n$ = 7[b]) | 25 mg/kg | 14.1 (25) | 4.9 (21) | 1.3 (12) | 21.9 (24) | |
| Intravenous acyclovir | No data | No data | No data | No data | No data | |
| 12 through 23 months | | | | | | 47 |
| Oral valacyclovir ($n$ = 6) | 10 mg/kg | 6.6 (27) | 2.5 (37) | 1.4 (22) | 18.8 (23) | |
| Oral valacyclovir ($n$ = 6) | 20 mg/kg | 14.4 (43) | 4.0 (34) | 2.1 (41) | 18.9 (41) | |
| Oral valacyclovir ($n$ = 9[c]) | 25 mg/kg | 15.3 (34) | 4.7 (44) | 1.3 (7) | 21.2 (38) | |
| Intravenous acyclovir ($n$ = 6) [27, 28] | 14.8 mg/kg[a] | 27.3 (63) | 13.0 (46) | 1.9 (18) | 9.3 (30) | |
| 2 through 5 years | | | | | | 48 |
| Oral valacyclovir ($n$ = 12[d]) | 10 mg/kg | 5.7 (30) | 2.1 (23) | 1.7 (45) | 21.9 (31) | |
| Oral valacyclovir ($n$ = 12[e]) | 20 mg/kg | 10.1 (33) | 3.8 (30) | 1.5 (23) | 25.5 (35) | |
| Oral valacyclovir ($n$ = 12) | 25 mg/kg | 19.9 (13) | 6.5 (19) | 1.5 (14) | 14.9 (13) | |
| Intravenous acyclovir ($n$ = 11) [27–29] | 13.0 mg/kg[a] | 23.2 (47) | 12.8 (47) | 1.8 (32) | 10.0 (23) | |
| 6 through 11 years | | | | | | 51 |
| Oral valacyclovir ($n$ = 8) | 10 mg/kg | 8.8 (17) | 3.1 (15) | 1.6 (19) | 13.5 (17) | |
| Oral valacyclovir ($n$ = 8) | 20 mg/kg | 13.1 (26) | 4.7 (25) | 1.4 (11) | 18.7 (21) | |
| Intravenous acyclovir ($n$ = 17) [27–29] | 11.6 mg/kg[a] | 25.0 (54) | 12.1 (56) | 2.2 (42) | 8.2 (27) | |

**NOTE.** %CV, percent coefficient of variation; $AUC_{0-\infty}$, area under the concentration-time curve from time zero extrapolated to infinity; CL/F, apparent oral clearance; $C_{max}$, maximum observed concentration; $t_{1/2}$, terminal elimination half-life; UD, unable to determine.

[a] Mean dose.
[b] $n$ = 7 for $C_{max}$; $n$ = 6 for $AUC_{0-\infty}$, $t_{1/2}$, and CL/F.
[c] $n$ = 9 for $C_{max}$; $n$ = 7 for $AUC_{0-\infty}$, $t_{1/2}$, and CL/F.
[d] $n$ = 12 for $C_{max}$; $n$ = 11 for $AUC_{0-\infty}$, $t_{1/2}$, and CL/F.
[e] $n$ = 12 for $C_{max}$; $n$ = 11 for $AUC_{0-\infty}$, $t_{1/2}$, and CL/F.



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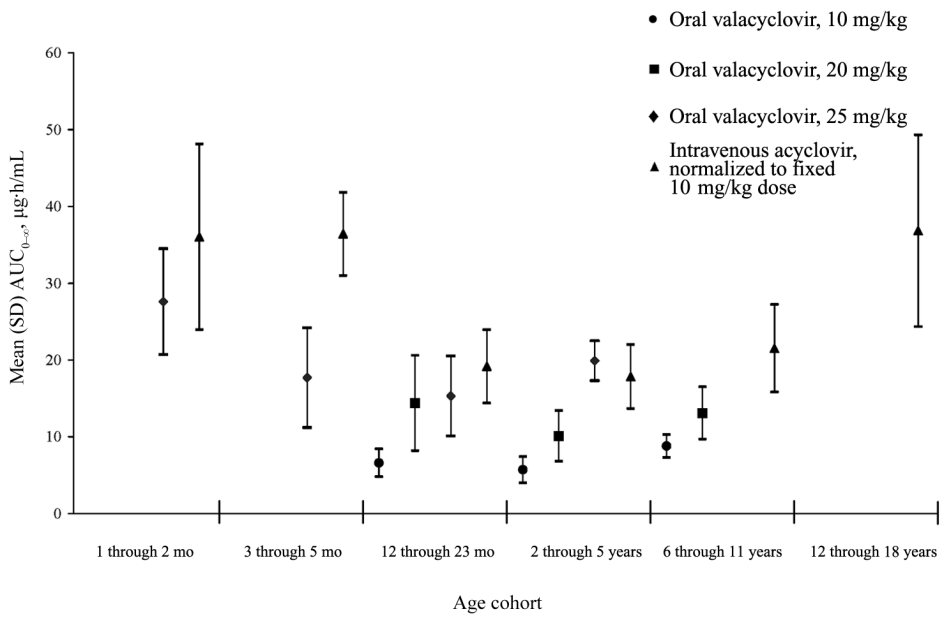



**Figure 1.** Comparison of mean acyclovir area under the concentration-time curve from time zero extrapolated to infinity ($AUC_{0-\infty}$), by formulation, dose, and age. Intravenous acyclovir values are those from Table 2 normalized to a 10 mg/kg dose.

in the older children. Bioavailability was estimated to be 45%–51% in all age groups except those 3 through 5 months old, in whom the bioavailability was lower at 22% (Table 2). Cross-study comparisons of mean results within age groups suggest that there is approximate dose proportionality in $C_{max}$ and $AUC_{0-\infty}$ across the 10 mg/kg to 25 mg/kg dose range (ie, dose-normalized differences generally within ~≤30%) with the exception of children in the 2- through 5-year age range, for whom a near doubling in $C_{max}$ and $AUC_{0-\infty}$ was noted with only a modest increase in dose from 20 to 25 mg/kg (Table 2 and Figures 1 and 2). Analysis of $AUC_{0-6}$ yielded the same results as that of $AUC_{0-\infty}$ (data not shown).

***Safety results.*** Across the 3 studies, 27 (24%) of 112 children reported adverse events: 12 patients (21%) in study 914, 9 patients (32%) in study 915, and 6 patients (22%) in study 916. All adverse events were mild or moderate in intensity. Eight (7%) of 112 children experienced an adverse event considered by the investigator to be drug related. Diarrhea was the only drug-related adverse event that occurred in >1 child (3 children). No adverse events were reported that led to premature discontinuation of either the study or use of the study drug.

Serious adverse events were reported in 3 (2.7%) of the 112 children across the 3 studies, with 1 child experiencing serious adverse events in each study: enteroviral meningitis in study 914; diarrhea, vomiting, and dehydration in study 915; and pneumonia in study 916. None of the serious adverse events was considered by the investigator to be drug related. There were no deaths or reports of thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, or TMA.

***Laboratory safety evaluations.*** Across all 3 studies, no clinically significant trends were noted in clinical chemical, hematologic, or urinalysis values from screening to follow-up. There was no laboratory evidence of TMA.

## DISCUSSION

Valacyclovir oral suspension at doses of 10 and 20 mg/kg administered twice daily to 3 times daily was well tolerated and in children 3 months through 11 years of age produced acyclovir $AUC_{0-\infty}$ and $C_{max}$ values similar to those for 500-mg and 1000-mg valacyclovir caplets, respectively, in adults. These results are similar to previous reports of other formulations of valacyclovir evaluated in pediatric patients [28, 30].

In children 2 through 5 years of age, the mean CL/F decreased from 25.5 to 14.9 mL/min/kg as the dose increased from 20 to 25 mg/kg, with corresponding increases in $C_{max}$ (3.8–6.5 $\mu$g/mL) and $AUC_{0-\infty}$ (10.1–19.9 $\mu$g·h/mL). The $AUC_{0-\infty}$ at the 25 mg/kg dose (19.9 $\mu$g·h/mL) more closely approximated that of 1000 mg in adults (18.8 $\mu$g·h/mL), suggesting that this might be the more appropriate dose for this age group, but as a single-dose study the 914 trial, in which the 25 mg/kg dose was evaluated, was not designed to assess multidose safety and tolerability, limiting our ability to confidently recommend this dose

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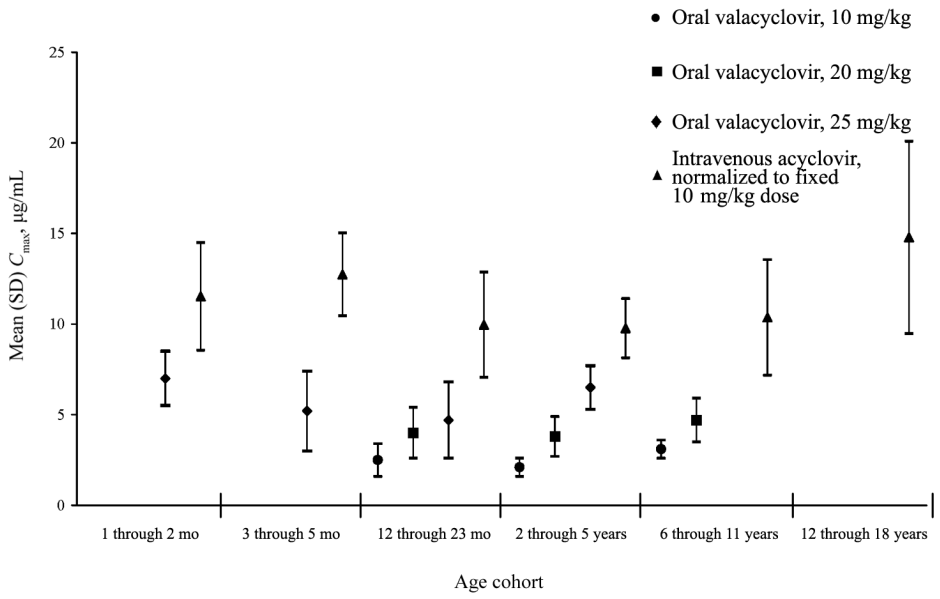



**Figure 2.** Comparison of mean acyclovir maximum observed concentration ($C_{max}$), by formulation, dose, and age. Intravenous acyclovir values are those from Table 2 normalized to a 10 mg/kg dose.

for young children. Acyclovir $AUC_{0-\infty}$ after a single dose was equivalent to the acyclovir AUC over a dosing interval at multiple-dose steady state. Acyclovir CL/F was derived from the $AUC_{0-\infty}$ by standard noncompartmental methods, which are not subject to any modeling assumptions.

Acyclovir exposure in infants 1 through 2 months of age was appreciably higher than in children 3 months through 11 years of age. Bioavailability in the 1- through 2-month age group was similar to that of older children compared with the 3- through 5-month age group, in whom bioavailability was appreciably lower. However, clearance was lower in the very young infants relative to that of children 3 months through 5 years of age, explaining the higher $AUC_{0-\infty}$ in that group. Differences in bioavailability by age with another nucleoside prodrug, valganciclovir, has been reported in young infants [31], and changes in expression of intestinal PEPT1 transporter in rats has been described [32], which could potentially explain the findings in this study given that the rat is a good model for drug absorption in humans [33, 34]. Although multiple-dose schedules of 10 and 20 mg/kg were assessed in these studies, repetitive dosing at 25 mg/kg was not evaluated; therefore, we do not have safety and tolerability data in the pediatric population for this highest dose of valacyclovir.

Given the lack of multidose safety and tolerability data for the 25 mg/kg dose and the higher acyclovir exposure with the modest increase in dose from 20 to 25 mg/kg, we recommend a valacyclovir oral suspension dose of 20 mg/kg administered twice daily or 3 times daily for children 3 months through 11 years of age, using the oral suspension formulation of the drug as detailed in the US prescribing information for valacyclovir [24]. The maximum dosage should not exceed 1000 mg/dose. Because VZV is relatively less sensitive to acyclovir than is HSV [21], we recommend 3-times-daily dosing for VZV infections and twice-daily dosing for HSV infections. The Food and Drug Administration recently approved valacyclovir oral suspension at the 20 mg/kg 3-times-daily dose for 5 days for the treatment of chickenpox in immunocompetent children 2 through 17 years of age [24].

Only 9 of the 12 very young infants 1 through 2 months of age studied in the 914 trial had pharmacokinetic data; all received 25 mg/kg as a single dose, and no pharmacokinetic data exist for lower valacyclovir doses in this age group. Therefore, no dosing recommendations can be made for valacyclovir oral suspension in infants <3 months of age. Oral acyclovir suspension has been better studied in this age group and should be considered first in infants <3 months of age who warrant oral (as opposed to intravenous) therapy. Further study of valacyclovir safety and pharmacokinetics in these very young infants is warranted.

Given valacyclovir's favorable dosing intervals compared with oral acyclovir, its use can be considered in the treatment of gingivostomatitis (HSV) [15], chickenpox (VZV) [16, 17], or herpes zoster (VZV) [18, 19] in immunocompetent children. Valacyclovir also can be considered in children for the treatment or suppression of herpes labialis (HSV) [14, 35–37]. In adults, clinical outcomes generally are comparable between oral acyclovir and valacyclovir, but the more favorable pharmacokinetics of the latter make it preferable for use in immunocompromised hosts, including children. Life-threatening herpesvirus infections should be treated with intravenous acy-



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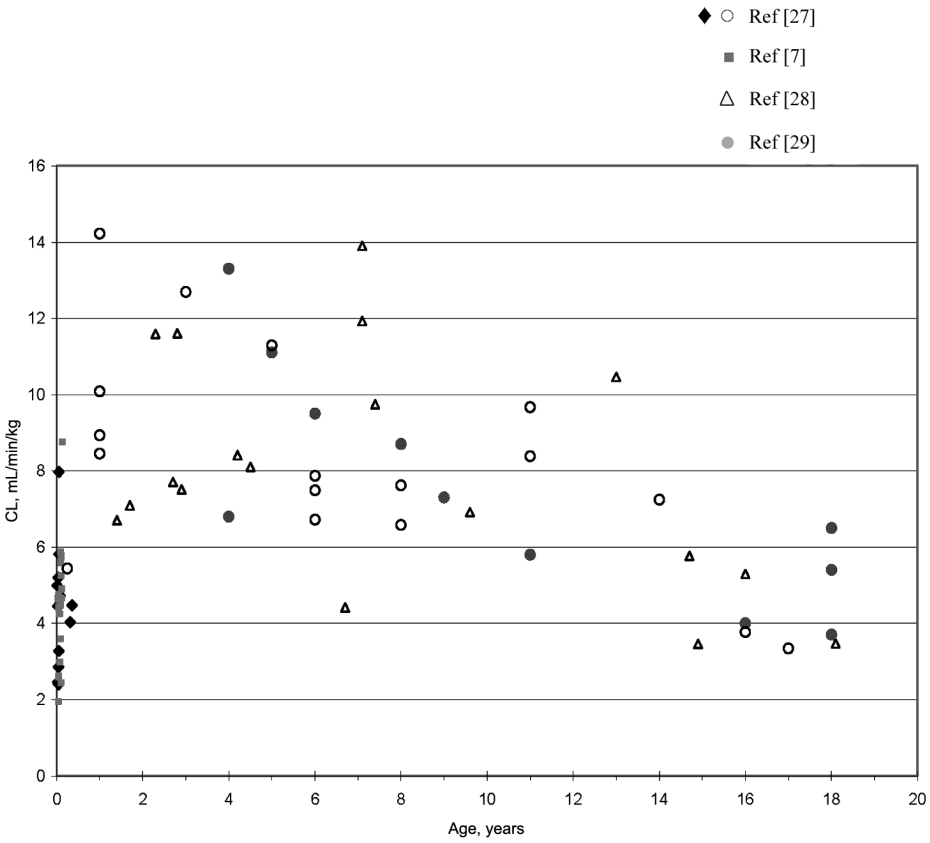



**Figure 3.** Intravenous acyclovir systemic clearance (CL) versus age.

clovir rather than oral antiviral therapy [7–13, 28], at least for the initial treatment period.

Oral acyclovir suppression after treatment of neonatal HSV disease has been investigated in a phase 1/2 study conducted by the National Institute of Allergy and Infectious Diseases Collaborative Antiviral Study Group (CASG) [38]. Data from 2 subsequent phase 3 placebo-controlled CASG trials are being analyzed at this time, with results anticipated in 2010. Currently, there is no evidence of benefit of suppressive therapy in this population, and there is a risk of adverse events (neutropenia and masked recurrence of disease) [38, 39]. As noted here, clearance of acyclovir after administration of valacyclovir to infants <3 months of age is diminished relative to that of older children. Therefore, valacyclovir is not preferable to oral acyclovir for suppressive therapy in infants after neonatal HSV disease, and indeed antiviral suppression in general in this circumstance is unproven at this time.

Overall, the safety profile observed in these 3 studies of short-term dosing of valacyclovir oral suspension was favorable, with few drug-related adverse events or laboratory abnormalities, which was not unexpected given the safety profile of acyclovir in children and valacyclovir in adults. Given the higher bioavailability and decreased dosing frequency compared with oral acyclovir, extemporaneous valacyclovir may present a more suitable option than oral acyclovir for the treatment of VZV and HSV infections in infants and children.

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